

CHY1001 – Engineering Chemistry

Module 1 & Module 2

(PYQs list and reference for Even sem MTE 2026 syllabus)

By Akshit Raj

🌟 <https://www.muji.stella.in/#/notes/study-materials/btech-1st-year>

Note: This is just a reference list. Refer to original PDFs mentioned for questions on the above website. As some might be missing some data, values or diagram/Graphs.

MODULE 1: WATER TREATMENT TECHNOLOGY

1. Hardness of Water – Concept

From: EC_ETE_Sem-I_2024-25.pdf

The hardness of water is due to

- (a) K and Na salts
- (b) Salts of iron
- (c) Salts of Ca and Mg
- (d) SiO_2

2. Determination of Hardness by EDTA Method

From: EC_ETE_Sem-I_2024-25.pdf

50 ml of standard hard water containing 1 gram of pure CaCO_3 per liter consumed 20 ml of EDTA. 50 ml of hard water consumed 25 ml of same EDTA solution and EBT indicator. Calculate the total hardness of water sample in ppm.

From: Solution -EC_ETE_Sem-I_2024-25.pdf

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From: EC_ETE_Sem-I_2025-26.pdf

Calculate the temporary hardness and permanent hardness of a sample of water containing:
 $\text{Mg}(\text{HCO}_3)_2 = 7.3 \text{ mg/L}$; $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}$; $\text{MgCl}_2 = 9.5 \text{ mg/L}$; $\text{CaSO}_4 = 13.6 \text{ mg/L}$.
[Atomic Mass: Ca = 40, Mg = 24, Na = 23, C = 12, O = 16, H = 1, Cl = 35.5, S = 32]

From: EC_MTE_Sem-I_2025-2026.pdf

50 ml of standard hard water containing 2 grams of pure CaCO_3 per liter consumed 20 ml of EDTA. 50 ml of hard water consumed 30 ml of same EDTA solution and EBT indicator. Calculate molarity of standard hard water and the total hardness (ppm) of water sample.

3. Internal Treatment of Hard Water

From: EC_ETE_Sem-I_2025-26.pdf

Explain the processes of internal treatment methods of hard water with chemical reactions.

From: EC_ETE_MakeupFEB2025.pdf

Explain internal conditioning methods of softening of hard water.

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4. Lime Soda Process (External Treatment)

From: EC_ETE_Solution April-May 2024.pdf

A sample of water on analysis gave the following results:

$\text{Ca}^{2+} = 30 \text{ mg/L}$; $\text{Mg}^{2+} = 18 \text{ mg/L}$; $\text{K}^+ = 19.5 \text{ mg/L}$; $\text{CO}_2 = 11 \text{ mg/L}$; $\text{HCO}_3^- = 122 \text{ mg/L}$; $\text{Na}^+ = 35.5 \text{ mg/L}$.

[Mol. wt.: $\text{Ca}^{2+} = 40$; $\text{Mg}^{2+} = 24$; $\text{K}^+ = 39$; $\text{CO}_2 = 44$; $\text{HCO}_3^- = 61$; $\text{Na}^+ = 23$]

(i) Calculate total hardness present in water sample.

(ii) Also calculate lime-soda required for softening of one litre of this sample of hard water.

From: EC_MTE_Sem-II_2023-24.pdf

A water sample contains the following impurities: $\text{Ca}^{2+} = 20 \text{ ppm}$, $\text{Mg}^{2+} = 18 \text{ ppm}$, $\text{HCO}_3^- = 183 \text{ ppm}$ and $\text{SO}_4^{2-} = 24 \text{ ppm}$. Calculate the amount of lime (90% Purity) and soda (80% purity) needed for softening of 10,000 liters of water.

[atomic/molecular weight of Calcium = 40, magnesium = 24, $\text{HCO}_3^- = 61$, $\text{SO}_4^{2-} = 96$]

From: EC_MTE_Sem-I_2025-2026.pdf

(i) Calculate the amount of lime (59% pure) and soda (80% pure) required for softening 50,000 liters of hard water with following composition. $\text{Ca}(\text{HCO}_3)_2 = 32.4 \text{ mg/L}$; $\text{Mg}(\text{HCO}_3)_2 = 146 \text{ mg/L}$; $\text{CaCl}_2 = 55.5 \text{ mg/L}$; $\text{MgSO}_4 = 30 \text{ mg/L}$; $\text{Na}_2\text{SO}_4 = 142 \text{ mg/L}$ and $\text{CaSO}_4 = 27.2 \text{ mg/L}$.

[Atomic Mass: Ca = 40, Mg = 24, Na = 23, C = 12, O = 16, H = 1, N = 14, Cl = 35.5, S = 32]

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5. Zeolite Process / Ion Exchange

From: EC_MTE_Sem-II_2023-24.pdf

With a neat diagram and reactions explain the zeolite process for the purification of water. Discuss demerits of this process also.

(Refer to original diagram in PDF)

From: EC_MTE_Re-sess_Sem-I_2025-26.pdf

(ii) A water sample containing CaCl_2 and MgSO_4 , write only reactions to remove these salts applying zeolite method.

From: EC_MTE_Sem-I_2025-2026.pdf

Identify a method for the preparation of deionized water from hardwater and describe the process.

MODULE 2: ADVANCED MATERIALS AND POLYMERS

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1. Polymer Basics

From: EC_MTE_Sem-I_2025-2026.pdf

What is the functionality of Terephthalic acid?

(1) Monofunctional (2) Bifunctional (3) Trifunctional (4) Polyfunctional

2. Degree of Polymerization / Molecular Weight

From: EC_ETE_Sem-I_2025-26.pdf

A polymer has a number-average molecular weight (M_n) of 30,000 g/mol. If the molecular weight of the repeating unit is 120 g/mol, the degree of polymerization (DP).

(1) 250 (2) 2500 (3) 360 (4) 3600

From: EC_ETE_Sem-I_2025-26.pdf

Consider a polymer sample containing 50 molecules of molecular weight 10000, 35 molecules of molecular weight 12000, and 15 molecules of molecular weight 14000. Determine number average molecular weight, weight average molecular weight. If the polymer is polypropylene, what will be the degree of polymerization?

From: EC_MTE_Sem-I_2025-2026.pdf

Find out the number average ($\bar{M}_n$) and weight average ($\bar{M}_w$) molecular weight of Teflon polymer with the composition as described below.

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$\text{F}_2\text{C}-\text{CF}_2$ (100) = 20% , $\text{F}_2\text{C}-\text{CF}_2$ (200) = 30 % and $\text{F}_2\text{C}-\text{CF}_2$ (250) = 50 %

[All % values are in number, atomic weights of fluorine (F) = 19, carbon (C)=12]

3. Free Radical Polymerization

From: EC_MTE_Sem-II_2024-25_solutions.pdf

Free radical polymerization

Mechanism with proper structures and steps

From: EC_ETE_MakeupFEB2025.pdf

Describe the role of benzoyl peroxide in addition polymerization of alkenes? Explain its mode of action with the help of reactions.

From: EC_ETE April-May 2024.pdf

Analyze mechanism in detail and draw tacticity of polymers obtained from vinyl chloride in presence of a peroxide.

(Refer to structure/tacticity diagrams in PDF)

4. Cationic Polymerization

From: EC_ETE_Sem-I_2024-25.pdf

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Note: This is just a reference list. Refer to original PDFs mentioned for questions on the above website. As some might be missing some data, values or diagram/Graphs.

Explain the mechanism of cationic polymerization reaction with suitable examples.

From: EC_MTE_Sem-I_2025-2026.pdf

Elaborate the polymerization mechanism of isobutene $[(CH_3)_2C=CH_2]$ in the presence of acid.

5. Anionic Polymerization

From: EC_ETE_Sem-I_2025-26.pdf

Describe the polymerization of vinyl chloride ($CH_2=CH-Cl$) in presence of potassium amide.

From: EC_MTE_Re-sess_Sem-I_2025-26.pdf

Elaborate the polymerization mechanism of acrylonitrile ($CH_2=CH-CN$) in the presence of n-Butyllithium.

6. Copolymerization

From: EC_MTE_Sem-II_2023-24.pdf

A co-polymer was synthesized by mixing 132.75 gram of 2-Chloro-buta-1,3-diene (MW= 88.5) and 31.25 gram vinyl chloride (MW=62.5). Calculate the ratio of the monomers in the polymer, also draw the structure of the repeating unit.

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(Refer to original repeating unit structure in PDF)

7. Nylon (6:6, 6:10)

From: EC_MTE_Sem-II_2023-24.pdf

Give the preparation and structure of the following:

3. Styrene-butadiene rubber

4. Nylon, 6:6

(Refer to original structure diagrams in PDF)

From: EC_ETE_Sem-I_2025-26.pdf

(ii) Synthesize the Nylon 6,10 using proper reagents and predict the final polymeric structure.

(Refer to structure diagram in PDF)

8. Vulcanization

From: EC_MTE_Sem-II_2024-25_solutions.pdf

Vulcanization of Rubber (Natural rubber+ Sulphur)